

# De Moivre's Theorem: Powers of Complex Numbers

Precalculus / Trigonometry Worksheet · Grade 11–12

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Learning Objectives

- Convert complex numbers from rectangular form to trigonometric (polar) form
- Apply De Moivre's Theorem to raise complex numbers to integer powers
- Evaluate trigonometric expressions using the unit circle and coterminal angles

## Problems

1. Find the modulus  $r$  of the complex number shown below.

$$z = 3 + 4i$$

2. Find the argument  $\theta$  (in degrees) of the complex number shown below. Round to the nearest degree if necessary.

$$z = 1 + \sqrt{3}i$$

3. Write the complex number shown below in trigonometric (polar) form  $z = r(\cos \theta + i \sin \theta)$ .

$$z = -1 + i$$

4. Use De Moivre's Theorem to evaluate the expression below. Write your answer in rectangular form  $a + bi$ .

$$[2(\cos 30^\circ + i \sin 30^\circ)]^3$$

5. Use De Moivre's Theorem to evaluate the expression below. Write your answer in rectangular form  $a + bi$ .

$$[3(\cos 45^\circ + i \sin 45^\circ)]^4$$



6. Convert the complex number shown below to trigonometric form, then use De Moivre's Theorem to compute the power. Write the final answer in rectangular form  $a + bi$ .

$$(1 + i)^6$$

7. Compute the power shown below using De Moivre's Theorem. Write the final answer in rectangular form  $a + bi$ .

$$(-1 + \sqrt{3}i)^5$$

8. Compute the power shown below using De Moivre's Theorem. Write the final answer in rectangular form  $a + bi$ .

$$(\sqrt{3} - i)^8$$

9. Use De Moivre's Theorem to compute the power shown below. Write the final answer in rectangular form  $a + bi$ .

$$(-1 + \sqrt{3}i)^{12}$$

10. Use De Moivre's Theorem to compute the power shown below. Write the final answer in rectangular form  $a + bi$ .

$$(2 + 2i)^{10}$$



# De Moivre's Theorem: Powers of Complex Numbers — Answer Key

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## Answer Key

### 1. Answer: $r = 5$

- Use  $r = \sqrt{a^2 + b^2}$  where  $a = 3$  and  $b = 4$
- $r = \sqrt{9 + 16} = \sqrt{25} = 5$

### 2. Answer: $\theta = 60^\circ$

- Use  $\theta = \tan^{-1}(b/a) = \tan^{-1}(\sqrt{3}/1)$
- $\tan^{-1}(\sqrt{3}) = 60^\circ$ , and since  $a > 0$  and  $b > 0$  the angle is in Quadrant I, so  $\theta = 60^\circ$

### 3. Answer: $z = \sqrt{2} (\cos 135^\circ + i \sin 135^\circ)$

- $r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$
- Reference angle:  $\tan^{-1}(1/1) = 45^\circ$ . Since  $a < 0$  and  $b > 0$ , the angle is in Quadrant II:  $\theta = 180^\circ - 45^\circ = 135^\circ$
- $z = \sqrt{2} (\cos 135^\circ + i \sin 135^\circ)$

### 4. Answer: $8i$ (i.e. $0 + 8i$ )

- Apply De Moivre's:  $r^n = 2^3 = 8$ , and  $n \cdot \theta = 3 \times 30^\circ = 90^\circ$
- Result:  $8(\cos 90^\circ + i \sin 90^\circ) = 8(0 + i \cdot 1) = 8i$

### 5. Answer: $-81 + 0i$

- $r^4 = 3^4 = 81$ , and  $n \cdot \theta = 4 \times 45^\circ = 180^\circ$
- $81(\cos 180^\circ + i \sin 180^\circ) = 81(-1 + 0i) = -81$

### 6. Answer: $-8 + 0i$

- $r = \sqrt{1^2 + 1^2} = \sqrt{2}$ ,  $\theta = 45^\circ$
- Trig form:  $\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$
- Apply De Moivre's:  $r^6 = (\sqrt{2})^6 = 8$ ,  $n \cdot \theta = 6 \times 45^\circ = 270^\circ$
- $8(\cos 270^\circ + i \sin 270^\circ) = 8(0 - i) = -8i$
- Wait —  $\cos 270^\circ = 0$  and  $\sin 270^\circ = -1$ , so answer  $= 8(0 + i(-1)) = -8i$

### 7. Answer: $16 + 16\sqrt{3}i$

- $a = -1$ ,  $b = \sqrt{3}$ ;  $r = \sqrt{1 + 3} = 2$
- Reference angle:  $\tan^{-1}(\sqrt{3}/1) = 60^\circ$ . Since Quadrant II:  $\theta = 120^\circ$
- Apply De Moivre's:  $r^5 = 2^5 = 32$ ,  $n \cdot \theta = 5 \times 120^\circ = 600^\circ$
- Coterminal angle of  $600^\circ$ :  $600^\circ - 360^\circ = 240^\circ$
- $32(\cos 240^\circ + i \sin 240^\circ) = 32(-1/2 + i(-\sqrt{3}/2)) = -16 - 16\sqrt{3}i$

### 8. Answer: $128 + (-128\sqrt{3})i \approx 128 - 128\sqrt{3}i$

- $a = \sqrt{3}$ ,  $b = -1$ ;  $r = \sqrt{3 + 1} = 2$





- $\theta = \tan^{-1}(-1/\sqrt{3}) = -30^\circ$  (Quadrant IV)  $= 330^\circ$
- $r^8 = 2^8 = 256$ ,  $n \cdot \theta = 8 \times 330^\circ = 2640^\circ$
- Coterminal:  $2640^\circ \bmod 360^\circ = 2640 - 7 \times 360 = 2640 - 2520 = 120^\circ$
- $256(\cos 120^\circ + i \sin 120^\circ) = 256(-1/2 + i\sqrt{3}/2) = -128 + 128\sqrt{3}i$

**9. Answer:  $4096 + 0i$** 

- $r = 2$ ,  $\theta = 120^\circ$  (from Quadrant II)
- $r^{12} = 2^{12} = 4096$ ,  $n \cdot \theta = 12 \times 120^\circ = 1440^\circ$
- Coterminal:  $1440^\circ - 4 \times 360^\circ = 1440^\circ - 1440^\circ = 0^\circ$
- $4096(\cos 0^\circ + i \sin 0^\circ) = 4096(1 + 0i) = 4096$

**10. Answer:  $0 - 32768i$** 

- $a = 2$ ,  $b = 2$ ;  $r = \sqrt{4+4} = 2\sqrt{2}$ ,  $\theta = \tan^{-1}(2/2) = 45^\circ$
- $r^{10} = (2\sqrt{2})^{10} = 2^{10} \cdot (\sqrt{2})^{10} = 1024 \cdot 32 = 32768$
- $n \cdot \theta = 10 \times 45^\circ = 450^\circ$
- Coterminal:  $450^\circ - 360^\circ = 90^\circ$
- $32768(\cos 90^\circ + i \sin 90^\circ) = 32768(0 + i \cdot 1) = 32768i$

